Certification of Imitation Controllers Using Learned Lyapunov Functions

Master thesis by Josua Christoph Lindemann
Date: July 2, 2026, Date of submission: 16. August 2026

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Darmstadt

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Contents

1 Abstract 1

2 Introduction 2

3 Optimal Control and Lyapunov Stability Theory 3
- 3.1 Discrete-Time Dynamical Systems and Stability . . . 3
  - 3.1.1 Lyapunov Stability . . . 3
- 3.2 Model Predictive Control and Stability . . . 4
  - 3.2.1 Problem Formulation . . . 4
  - 3.2.2 Terminal Ingredients for Stability . . . 4
  - 3.2.3 Closed-Loop Stability . . . 6
- 3.3 Stability without Terminal Constraints . . . 6

4 Foundations of Neural Control Policies 7
- 4.1 Artificial Neural Networks . . . 7
  - 4.1.1 Multi-Layer Perceptrons . . . 7
  - 4.1.2 Optimization and Training . . . 7
- 4.2 Counterexample-Guided Inductive Synthesis . . . 7

5 Formal Verification of Neural Control Policies 8
- 5.1 Mixed-Integer Linear Programming . . . 8
- 5.2 Branch and Bound . . . 8
- 5.3 $\alpha\beta$-CROWN . . . 8

6 Learning Lyapunov-Stable Imitation Controllers 9
- 6.1 Neural Policy Approximation and Imitation Learning . . . 9
  - 6.1.1 Data Generation and MPC Expert . . . 9
  - 6.1.2 Policy Network Architecture and Imitation Objective . . . 9
- 6.2 Lyapunov Learning for Stability Certification . . . 9
  - 6.2.1 Lyapunov Network Architecture . . . 9
  - 6.2.2 Loss Formulation and CEGIS Training . . . 9
- 6.3 Formal Verification Methodology . . . 9

7 Experimental Evaluation and Certification Results 10
- 7.1 System Modeling and Problem Formulation . . . 10
  - 7.1.1 Double Integrator . . . 10
  - 7.1.2 Cartpole . . . 10
- 7.2 Linear System Results: Double Integrator . . . 11
- 7.3 Nonlinear System Results: Cartpole . . . 11

8 Discussion **12**
8.1 Qualitative Discussion of Results 12
8.2 Challenges with Nonlinear Dynamics 12
8.3 Scalability and Limitations 12

9 Conclusion **13**

A Appendix **14**

1 Abstract

2 Introduction

3 Optimal Control and Lyapunov Stability Theory

3.1 Discrete-Time Dynamical Systems and Stability

To implement an optimal control scheme on digital hardware, the underlying continuous-time system dynamics

$$\dot{x}(t) = f(x(t), u(t)) \tag{3.1}$$

must be discretized, where $x(t) \in \mathcal{X}$ and $u(t) \in \mathcal{U}$ denote the continuous-time state and input vectors, respectively. In this work, the corresponding discrete-time system

$$x_{k+1} = f_d(x_k, u_k) \tag{3.2}$$

is obtained by approximating the continuous dynamics using a numerical integration scheme with a constant sampling time T_s, such as the forward Euler or a fourth-order Runge-Kutta (*RK4*) method **something**.

3.1.1 Lyapunov Stability

Let the origin $(x, u) = (\mathbf{0}, \mathbf{0})$ be an equilibrium point of the system such that $f_d(\mathbf{0}, \mathbf{0}) = \mathbf{0}$. Suppose the system is controlled by a state-feedback control policy $u_k = \kappa(x_k)$. The closed-loop system is said to be *Lyapunov stable* if there exists a positive definite Lyapunov function $V(x_k)$ such that:

$$\begin{aligned} V(\mathbf{0}) &= 0 & \\ V(x_k) &> 0 & \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\} \\ V(f_d(x_k, \kappa(x_k))) - V(x_k) &\leq 0 & \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\}. \end{aligned} \tag{3.3}$$

Furthermore, the equilibrium point is *locally asymptotically stable* if the Lyapunov function additionally satisfies a strictly negative definite decrease condition along the closed-loop trajectories:

$$V(f_d(x_k, \kappa(x_k))) - V(x_k) < 0 \qquad \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\}. \tag{3.4}$$

Asymptotic stability is typically the desired property for control systems, as it ensures that the system will converge to the equilibrium point over time.

3.2 Model Predictive Control and Stability

This section provides a brief overview of the Model Predictive Control (*MPC*) framework and its stability properties. First, the general MPC problem is formulated, followed by a discussion on the terminal set and terminal cost constraints. Afterwards, the closed-loop stability of the MPC scheme is analyzed.

3.2.1 Problem Formulation

For the nominal MPC setup considered in this work, the following finite-horizon optimal control problem is solved at each time step k given the current system state x_k:

$$\begin{aligned} \min_{\{x_i\}_{i=0}^{N}, \{u_i\}_{i=0}^{N-1}} \quad & \sum_{i=0}^{N-1} \ell(x_i, u_i) + V_f(x_N) \\ \text{s.t.} \quad & x_{i+1} = f_d(x_i, u_i), && i = 0, \dots, N-1, \\ & x_i \in \mathcal{X}, && i = 0, \dots, N, \\ & u_i \in \mathcal{U}, && i = 0, \dots, N-1, \\ & x_0 = x_k. \end{aligned} \tag{3.5}$$

Here, N denotes the prediction horizon, $\ell(x_i, u_i)$ is the stage cost function, and $V_f(x_N)$ represents the terminal cost function. The system dynamics are represented by the discrete-time function $f_d(x_i, u_i)$, while $\mathcal{X}$ and $\mathcal{U}$ denote the state and input constraint sets, respectively.

A quadratic cost function is used for the stage cost, defined as

$$\ell(x_i, u_i) = x_i^\top Q x_i + u_i^\top R u_i, \tag{3.6}$$

where $Q \succeq 0$ and $R \succ 0$ are the state and input weighting matrices, respectively. Following the receding horizon principle, only the first control input u_0^* from the optimal sequence is applied to the system, and the optimization is repeated at the next sampling instance.

3.2.2 Terminal Ingredients for Stability

To ensure stability of a closed-loop system of a nominal MPC, a terminal set $\mathcal{X}_f \subseteq \mathcal{X}$ and a terminal cost function $V_f(x_N)$ are introduced. The terminal set $\mathcal{X}_f$ is a positively invariant set for the system under a local control law $u = \mu(x_k)$, and the terminal cost $V_f(x_N)$ is a Lyapunov function for the system within this set. However, to include these terminal ingredients, Eq. (3.5) must be augmented with the following terminal constraint:

$$x_N \in \mathcal{X}_f. \tag{3.7}$$

The property of a positive invariant set $\mathcal{X}_f$ is defined as a set of states where a state $x_k \in \mathcal{X}_f$ remains in $\mathcal{X}_f$ for all future time steps under the local control law $\mu(x_k)$,

$$f_d(x, \mu(x)) \in \mathcal{X}_f \quad \forall x \in \mathcal{X}_f. \tag{3.8}$$

This predictive mechanism is visualized in Fig. 3.1. While the initial state x_0 and the intermediate predicted states x_1, x_2 are only restricted to the safe state space $\mathcal{X}$, the terminal constraint Eq. (3.7) forces the final predicted state x_N to terminate strictly inside the invariant terminal set $\mathcal{X}_f$. Together with the terminal cost $V_f(x_N)$, which acts as a local Lyapunov function inside $\mathcal{X}_f$, these ingredients establish a mathematical bridge.

For nonlinear systems, the terminal set $\mathcal{X}_f$ is typically computed using linearized system dynamics around the equilibrium point $(x, u) = (\mathbf{0}, \mathbf{0})$, yielding the linear discrete-time state-space model:

$$x_{k+1} = Ax_k + Bu_k. \tag{3.9}$$

A widely used choice for these terminal ingredients is then derived from the Linear-Quadratic Regulator (*LQR*). Here, the local control law $\mu(x_k)$ is chosen as a linear state-feedback policy $\mu(x_k) = -Kx_k$, where the optimal LQR gain K is given by

$$K = \left(R + B^\top PB\right)^{-1} B^\top PA. \tag{3.10}$$

The matrix $P \succ 0$ is obtained as the unique positive definite solution to the discrete-time Algebraic Riccati Equation (*DARE*):

$$P = A^\top PA - A^\top PB \left(R + B^\top PB\right)^{-1} B^\top PA + Q. \tag{3.11}$$

This matrix parameterizes the quadratic terminal cost function $V_f(x_N) = x_N^\top P x_N$. Accordingly, the terminal set $\mathcal{X}_f$ is defined as an ellipsoidal sub-level set of this cost function:

$$\mathcal{X}_f = \left\{ x \in \mathbb{R}^n \;\middle|\; x^\top P x \leq \alpha \right\}. \tag{3.12}$$

To satisfy the positive invariance condition Eq. (3.8) for the original nonlinear system, the scaling parameter $\alpha > 0$ must be chosen sufficiently small. This ensures not only that all states within $\mathcal{X}_f$ satisfy the state constraints $\mathcal{X}$ and input constraints $\mathcal{U}$, but also that the local quadratic Lyapunov descent condition dominates the higher-order linearization errors of the nonlinear dynamics.

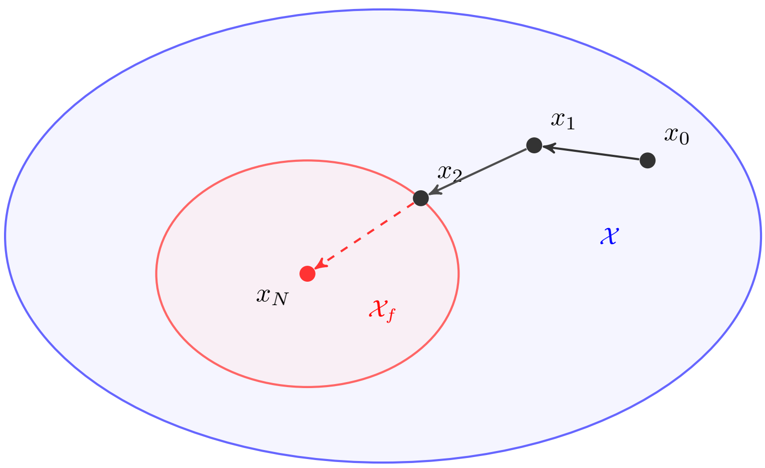

Figure 3.1: Comparison of the terminal set $\mathcal{X}_f$ (red) with the entire state space $\mathcal{X}$ (blue).

3.2.3 Closed-Loop Stability

TODO: ADD FORMULATION FOR RELAXED MPC STABILITY CONDITION

$$V_N(f_d(x_k, \mu(x_k))) \leq V_N(x_k) - \alpha_N \cdot \ell(x_k, \mu(x_k)), \quad \forall x_k \in \mathcal{X}_f \tag{3.13}$$

3.3 Stability without Terminal Constraints

only for no terminal constraints -> not asymptotically stable, but can be shown to be practically stable

$$\alpha_N = 1 - \frac{(\gamma - 1)^N}{\gamma^{N-1} - (\gamma - 1)^N} \tag{3.14}$$

if $\alpha_N > 0$ then

$$N < 2 + \frac{\ln(\gamma - 1)}{\ln(\gamma) - \ln(\gamma - 1)}, \tag{3.15}$$

then holds

$$V_N(x_k) \leq \gamma \ell^*(x_k) \tag{3.16}$$

4 Foundations of Neural Control Policies

4.1 Artificial Neural Networks

4.1.1 Multi-Layer Perceptrons

4.1.2 Optimization and Training

4.2 Counterexample-Guided Inductive Synthesis

5 Formal Verification of Neural Control Policies

5.1 Mixed-Integer Linear Programming

5.2 Branch and Bound

5.3 $\alpha\beta$-CROWN

6 Learning Lyapunov-Stable Imitation Controllers

6.1 Neural Policy Approximation and Imitation Learning

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6.2 Lyapunov Learning for Stability Certification

6.2.1 Lyapunov Network Architecture

6.2.2 Loss Formulation and CEGIS Training

6.3 Formal Verification Methodology

7 Experimental Evaluation and Certification Results

7.1 System Modeling and Problem Formulation

For our experimental setup, two benchmark systems are selected: the double integrator and the cartpole. These systems provide a good balance between simplicity and complexity allowing us to certify the stability of the closed-loop system.

7.1.1 Double Integrator

The double integrator is a simple second-order linear system; therefore, it serves as the ideal baseline system in the learning and certification pipeline. Discretizing the continuous-time dynamics using the explicit Euler method with a sampling time of $\Delta t = 0.1$, leads to the discrete-time state-space equation:

$$x_{k+1} = f_{\mathrm{DI},d}(x_k, u_k) = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ \Delta t \end{bmatrix} u_k \tag{7.1}$$

where $x_k = \begin{bmatrix} p_k & v_k \end{bmatrix}^\top$ is the state vector containing the position and velocity of the system, respectively, and u_k denotes acceleration.

The *MPC setup* is formulated according to the problem in Eq. (3.5) subject to the double-integrator dynamics in Eq. (7.1). A prediction horizon of $N = 20$ steps is chosen, and the states and inputs are penalized using the weighting matrices $Q = \mathrm{diag}(15, 1)$ and $R = 0.1$.

7.1.2 Cartpole

The inverted pendulum on a cart, commonly referred to as the cartpole system, is used as a nonlinear benchmark system in this work. There are four states in the system: the cart position p, the cart velocity v, the pendulum angle θ, and the pendulum angular velocity ω. Here, the state vector is defined as $x = \begin{bmatrix} p & v & \theta & \omega \end{bmatrix}^\top$, and the input u is the force F applied to the cart. This leads to the following continuous-time dynamics of the cartpole system:

$$f_{\text{CP}}(x,u) = \begin{bmatrix} \dot{p} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v \\ \frac{F + ml\omega^2 \sin(\theta) - mg \sin(\theta)\cos(\theta)}{M + m - m\cos^2(\theta)} \\ \omega \\ \frac{-F\cos(\theta) - ml\omega^2 \sin(\theta)\cos(\theta) + (M+m)g\sin(\theta)}{l(M + m - m\cos^2(\theta))} \end{bmatrix} \tag{7.2}$$

where $M = 1\,\text{kg}$ is the mass of the cart, $m = 0.1\,\text{kg}$ is the mass of the pendulum, $l = 0.8\,\text{m}$ is the length of the pendulum, and $g = 9.81\,\text{m/s}^2$ is the acceleration due to gravity. Discretizing the continuous-time dynamics using the explicit Euler method with a sampling time of $\Delta t = 0.05\,\text{s}$, leads to the discrete-time state-space equation:

$$x_{k+1} = f_{\text{CP},d}(x_k, u_k) = x_k + \Delta t f_{\text{CP}}(x_k, u_k) \tag{7.3}$$

For the nonlinear benchmark, a prediction horizon of $N = 40$ stages is chosen. The system states and control inputs are penalized within the optimization framework of Eq. (3.5) using the weighting matrices $Q = \text{diag}(10, 1, 10, 0.01)$ and $R = 0.5$, respectively, subject to the dynamics defined in Eq. (7.3).

7.2 Linear System Results: Double Integrator

7.3 Nonlinear System Results: Cartpole

8 Discussion

8.1 Qualitative Discussion of Results

8.2 Challenges with Nonlinear Dynamics

8.3 Scalability and Limitations

9 Conclusion

A Appendix